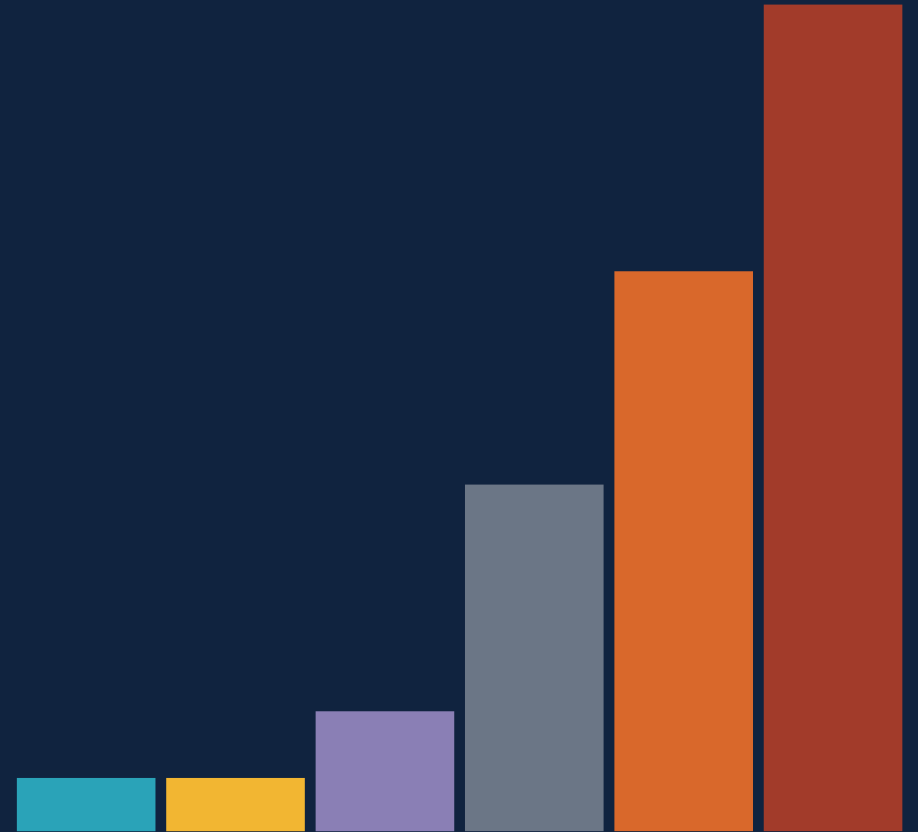


WORKSHOP · 50 MINUTES

European electricity markets

How wind and sun set the price of power

With two group tasks · As of October 2026



OVERVIEW

The agenda in 50 minutes

Time	Phase	Content
0–5 min	Warm-up	Guessing question on negative prices
5–16 min	Input 1	Market, trading venues, merit order
16–26 min	Group task 1	Build and solve your own merit order
26–33 min	Input 2	Renewables in the market, limits, outlook
33–47 min	Group task 2	Role play: one Sunday, two prices
47–50 min	Wrap-up	Three key messages

Yellow: this is where you work in groups.

GUESSING QUESTION

For how many hours in 2025 did electricity in Germany cost less than €0?

A

0

hours

B

50

hours

C

570

hours

D

2,000

hours

Show of hands: who says A, B, C or D? The answer comes later.

One market for 27 countries

The EU electricity market is the largest integrated electricity market in the world.

Always in balance

Electricity can hardly be stored.
Generation and consumption must
match every second.

Bidding zones

Each zone has one price.
Usually a zone is a country;
Germany and Luxembourg
share one.

Market coupling

Power flows from the cheap zone to the
expensive one until the lines at the
border are full.

Four markets before delivery

Years to days ahead

Forwards

Lock in prices long term, for example with supply contracts for wind and solar farms.

The day before

Day-ahead

One auction for all of Europe sets the price for every quarter-hour.

Right up to delivery

Intraday

Trade again when the forecast for wind or sun changes.

In real time

Balancing

The grid operators even out the last deviations.

The closer to delivery, the better the weather forecast. That helps wind and solar.

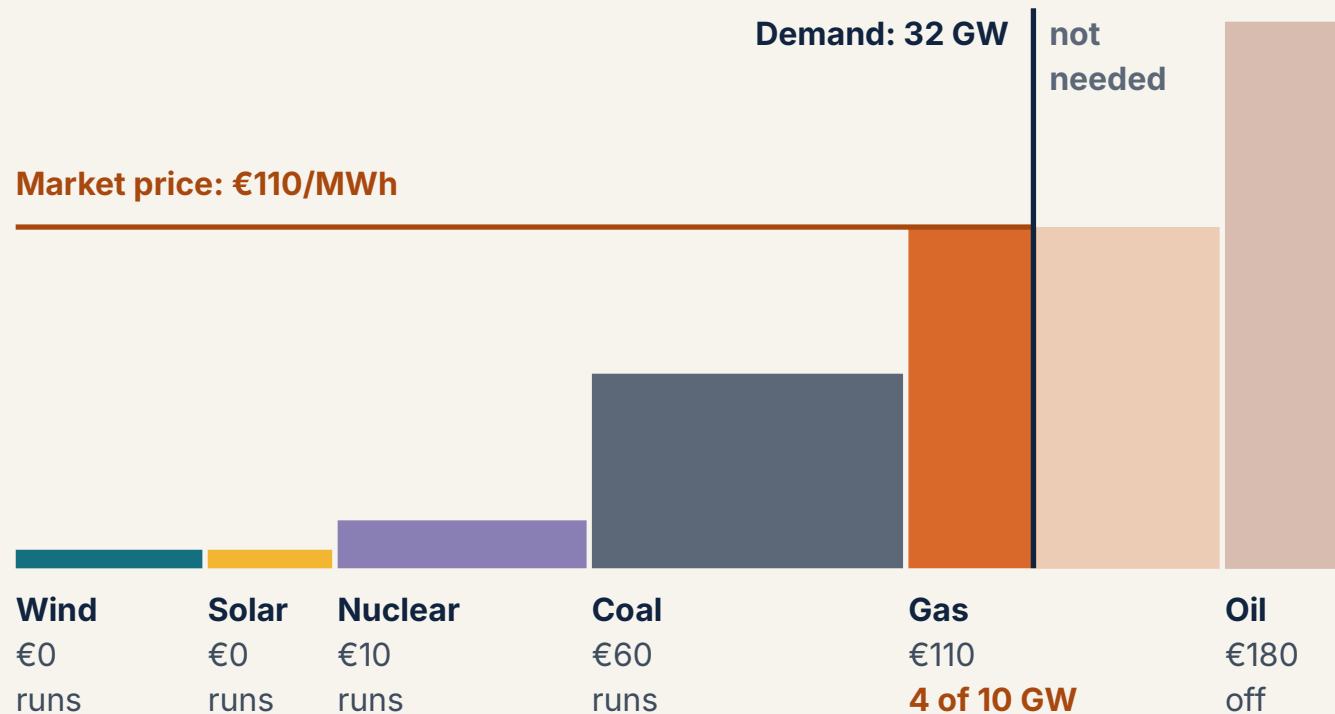
How the electricity price is set

Example: At 7 pm, 32 GW are needed. Bids are sorted from cheap to expensive.

Power plant	offers	at a price of	in total	Result
Wind	6 GW	€0/MWh	6 GW	runs
Solar	4 GW	€0/MWh	10 GW	runs
Nuclear	8 GW	€10/MWh	18 GW	runs
Coal	10 GW	€60/MWh	28 GW	runs
Gas	10 GW	€110/MWh	38 GW	sets the price: €110
Oil	4 GW	€180/MWh	42 GW	does not run

With coal, only 28 GW are reached. Gas supplies the missing 4 GW. Simplified example with invented values.

As a picture: the price staircase



How to read it: Each step is a power plant. Wide means lots of capacity, tall means expensive.

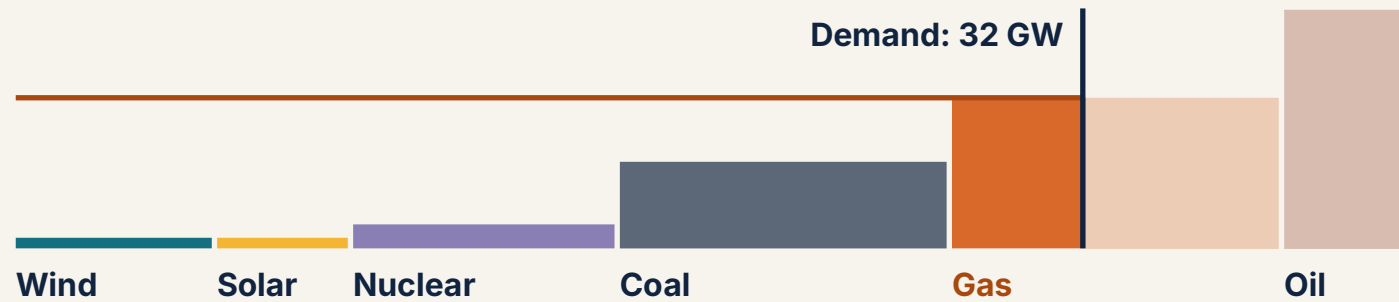
The demand line hits the gas step. So power now costs €110/MWh.

Everyone left of the line receives €110. Even the wind farm, which asked for €0.

INPUT 1 · THE MERIT ORDER, STEP 3

More wind, lower price

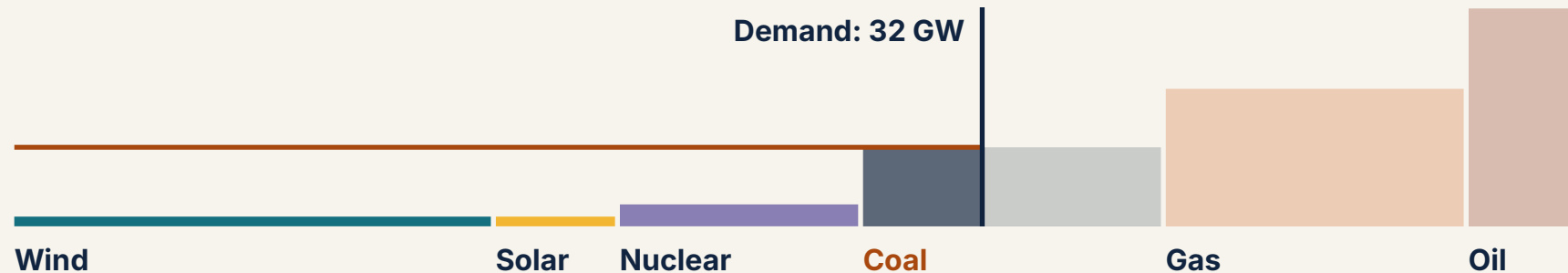
Little wind: 6 GW



€110

per MWh. Gas sets the price.

Lots of wind: 16 GW



€60

per MWh. Coal sets the price.

Same demand, same power plants, just 10 GW more wind. Simplified example with invented values.

GROUP TASK 1 · 6 MINUTES

Build your own merit order

Power plant	Capacity	Marginal cost
Wind	up to 12 GW	€0/MWh
Solar	up to 8 GW	€0/MWh
Nuclear	8 GW	€10/MWh
Coal	10 GW	€60/MWh
Gas	10 GW	€110/MWh
Oil	4 GW	€180/MWh

A · Calm winter evening

Wind 2 GW, solar 0 GW, demand 26 GW

B · Windy summer midday

Wind 12 GW, solar 8 GW, demand 26 GW

C · Sunny, windy Sunday

Wind 12 GW, solar 8 GW, demand 18 GW

- 1. Which power plant sets the price?**
- 2. How high is the market price?**
- 3. What happens to the surplus in C?**

GROUP TASK 1 · SOLUTION

Who sets the price?

Scenario	Wind + solar	Price setter	Market price
A · Winter evening	2 GW	Gas	€110/MWh
B · Summer midday	20 GW	Nuclear	€10/MWh
C · Sunday	20 GW (2 GW surplus)	Wind / solar	€0 or negative

Same demand, €100 difference. More wind and solar push the expensive plants out of the merit order.

One price for all. In A the wind farm also receives €110/MWh. That makes cheap generation attractive.

Renewables in the EU power mix 2025

30.1%

Wind and solar combined. For the first time more than all fossil plants (29.0%).

47.7%

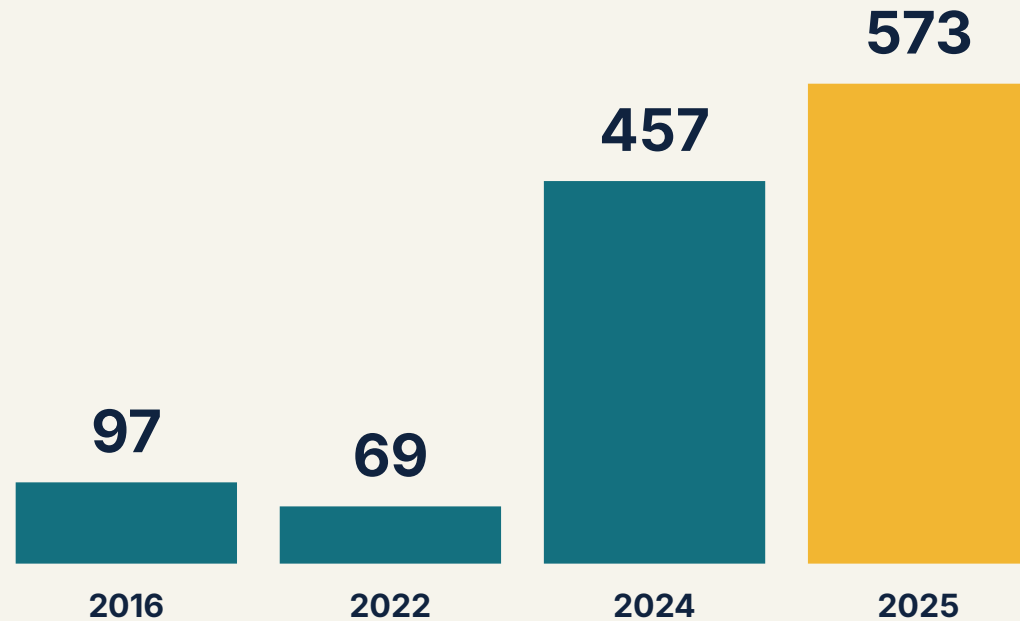
All renewables together: wind, solar, hydro and biomass.

14

of 27 EU countries generated more power from wind and solar than from fossil sources.

Solar grew by about 20% to 369 TWh and supplied 13% of EU electricity. Wind supplied 17%.

Electricity for less than zero euros



Hours with a negative day-ahead price in Germany

C was right: 573 hours, a record.

- Lots of wind and sun meet low demand.
- Some power plants cannot be shut down quickly.
- Storage and flexible consumers are still missing.

Not just Germany: in 2025 six European countries exceeded 500 hours.

Benefits of the common market

Weather evens out

Calm in the north, sun in the south: across many countries, wind and solar fluctuate less.

Less curtailment

Surplus wind and solar power flows to the neighbouring country instead of going unused.

Cheapest first

The merit order works across borders. Expensive plants run less often.

More security

If a power plant or a line fails, the neighbours step in.

Where the market hits its limits

Gas often sets the price

Q2 2026 in Germany: €95.21/MWh, 36.5% more than a year earlier. The main reason was more expensive gas.

Lines are scarce

Q2 2026: France €53/MWh, Germany €95/MWh. Full lines keep the prices apart.

Flexibility is missing

Storage, controllable consumers and dynamic tariffs must grow along with wind and solar.

What if all power were renewable?

The price

- Often near €0: whenever wind and sun cover the whole demand.
- Otherwise storage, hydro, biomass or hydrogen plants set the price.
- Consumers matter too: the price shows what they are willing to pay.

The pay

- On the market: in all hours with a price above zero.
- Long-term contracts with companies (PPA) or with the state (CfD).
- Payments for staying available, for storage and backup plants.

Model study: zero prices in about 30% of hours if consumers react to prices, about 90% if they do not.

GROUP TASK 2 · THE SITUATION AND YOUR JOB

One Sunday, two prices

1 pm · power in abundance

–€50/MWh

Whoever takes power gets paid. Whoever feeds in has to pay.

7 pm · power is scarce

€180/MWh

The sun is gone, everyone needs power. Every MWh is expensive.

1. Each group takes on a role.
2. Discuss for 5 minutes: what do you do at 1 pm, and at 7 pm?
3. Work out what that earns or costs you.
4. Present your result in 1 minute.

Tip: taking 10 MWh at –€50/MWh earns you €500.

GROUP TASK 2 · 5 MINUTES + PRESENTATION

Your roles

Role	What you have	What you decide
Solar farm	generates 100 MWh per hour	feed in or switch off?
Battery storage	holds 10 MWh, empty right now	when to charge and discharge?
Steelworks	needs 50 MWh per hour	when does the furnace run?
Household	electric car needs 40 kWh	when is it charged?
Grid operator	2 GW too much power in the grid	where does the surplus go?

How to present: "We are ... At 1 pm ... At 7 pm ... That earns us ..."

GROUP TASK 2 · SOLUTION

Everyone follows the same price

Role	1 pm · –€50/MWh	7 pm · €180/MWh
Solar farm	switch off, saves €5,000	delivers nothing now
Battery storage	charge, receives €500	discharge, earns €1,800
Steelworks	produce, receives €2,500	cut back, saves €9,000
Household	charge the car, receives €2	don't charge, saves €7.20
Grid operator	export, curtail if needed	keep reserves ready

Use and store more at midday, use less in the evening: that is how the surplus disappears. This is called flexibility.

WRAP-UP

What you take away

- 1 The most expensive plant that is needed sets the price for everyone.
- 2 Wind and solar bid at almost zero and so push the price down.
- 3 The common market makes renewables cheaper and more secure, if grids and flexibility grow with them.

The key terms

Term	Meaning
Marginal cost	Cost of one additional MWh: mainly fuel and CO2 allowances
Merit order	Ranking of power plants by marginal cost, from cheap to expensive
Bidding zone	Area with one uniform exchange price, usually a country
Market coupling	Joint price calculation for all zones, limited by the lines
Day-ahead	Auction on the day before, bids close at 12 noon, prices per quarter-hour
Intraday	Continuous trading until shortly before delivery
Balancing energy	Reserve of the grid operators to balance within seconds to minutes
Curtailment	Wind or solar plants are throttled because the power is not needed
PPA	Long-term power supply contract between a generator and a buyer
Contract for difference	The state secures a minimum price and skims off extra revenue (CfD)

APPENDIX

Sources

- [Ember: European Electricity Review 2026](#) (EU power mix 2025)
- [European Commission: Electricity market design](#) (structure, reform, benefit of the internal market)
- [Bundesnetzagentur, SMARD: The electricity market in Q2 2026](#) (wholesale prices by country)
- [EPEX SPOT: Trading Brochure](#) (how the auction runs, price limits)
- [NEMO Committee: 15-Minute MTU in SDAC](#) (quarter-hour trading since 1 October 2025)
- [pV magazine, 5 January 2026](#) and [15 January 2026](#) (negative prices 2025, in German)
- [1KOMMA5°: Negativer Strompreis](#) (time series since 2016, in German)
- [Brown, Neumann and Riepin: Price formation without fuel costs](#) (Energy Economics, 2025; fully renewable system)
- [DGS: Der Jahresmarktwert Solar 2025](#) and [Stiftung Umweltenergierecht](#) (solar revenues, contracts for difference)
- [GTAI](#) (Switzerland–EU agreements, in German)

Example values without a source: the power plant data in both group tasks and in the merit order slides.